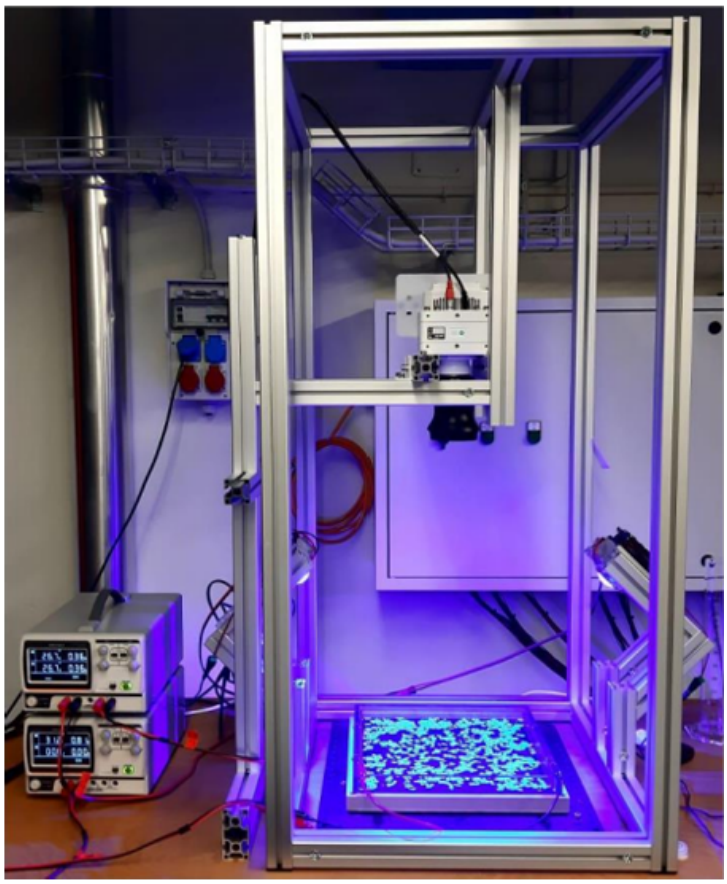
Experiment 1: Particle Image Velocimetry in a Thin Fluid Layer

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Course: Imaging Fluids
Experiment: Particle Image Velocimetry
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October 20, 2026

Abstract

This experiment investigates an electromagnetically driven quasi-two-dimensional flow using Particle Image Velocimetry. A thin layer of conductive $CuSO_4$ solution is placed above an 8×8 array of neodymium magnets with alternating polarity, and the flow is forced by applying an electric current through the fluid. Four current values are analysed, from 0.010 A to 0.500 A. The image sequences are processed with PIVlab to obtain velocity and vorticity fields. From these fields, the RMS velocity, Reynolds number, RMS vorticity, velocity fluctuation PDFs, kurtosis, and Eulerian velocity autocorrelation are computed. The measurements show a clear increase in flow intensity with current. The Reynolds number increases from 246 to 1711, while the RMS vorticity increases from 0.866 s^{-1} to 5.643 s^{-1}. The response is strong at low and intermediate forcing, but becomes weaker between the two highest currents, suggesting the beginning of saturation.

The spatial statistics show that the imposed magnet pattern remains visible, especially at low current. As the current increases, the autocorrelation oscillations weaken and the velocity PDFs move closer to a Gaussian distribution. The results therefore indicate a transition towards a less coherent quasi-two-dimensional flow, but not a fully developed turbulent regime over the full range of conditions.

1 Introduction

Thin fluid layers are useful for studying quasi-two-dimensional flows and are commonly used as model systems for two-dimensional turbulence [4, 5]. Because the fluid depth is small compared with the horizontal dimensions, vertical motion is limited and the flow mainly evolves in the horizontal plane. In this experiment, the flow is produced by electromagnetic forcing. A shallow layer of conducting $CuSO_4$ solution is placed above an array of permanent magnets. When an electric current passes through the fluid, the interaction between the current and the magnetic field produces a Lorentz force. This force drives the fluid and creates a regular forcing pattern linked to the magnet spacing. The goal of the experiment is to describe how the flow changes when the forcing is increased. Particle Image Velocimetry is used to measure the velocity field for four applied currents. From these measurements, several quantities are computed: the RMS velocity, Reynolds number, RMS vorticity, Eulerian velocity autocorrelation, velocity fluctuation PDFs, and kurtosis.

These measurements allow us to examine how the strength and spatial organisation of the flow change with increasing current. The driving mechanism is the Lorentz force per unit mass,

$$\mathbf{f}_L = \frac{\mathbf{J} \times \mathbf{B}}{\rho_f},$$

where $\mathbf{J}$ is the current density, $\mathbf{B}$ the magnetic field, and ρ_f the fluid density. For a fixed magnet arrangement, an approximate balance between this forcing and fluid inertia predicts $v_{\mathrm{RMS}} \propto \sqrt{I}$ and therefore $Re \propto \sqrt{I}$. This is a theoretical reference, not a relation imposed on the measured data [2].

2 Apparatus and Methodology

2.1 Experimental setup

The experiment was carried out in a rectangular tray with dimensions

$$L_1 \times L_2 = 0.300 \text{ m} \times 0.320 \text{ m}.$$

A thin layer of conductive $CuSO_4$ solution was poured into the tray. Two copper electrodes were placed on opposite sides in order to impose an electric current through the fluid. Below the tray, an 8×8 array of neodymium magnets with alternating polarity was used. Since the polarity changes from one magnet to the next, the electromagnetic forcing also changes direction. This generates a regular pattern of vortices in the fluid. The characteristic forcing length scale is the magnet spacing,

$$L_f = 3.5 \times 10^{-2} \text{ m}.$$

The fluid layer thickness was

$$h = 5.0 \times 10^{-3} \text{ m}.$$

This is much smaller than the horizontal dimensions of the tray, so the flow can be considered quasi-two-dimensional. Green fluorescent polyethylene particles were added as tracers. Their diameter was between 7.5×10^{-5} m and 9.0×10^{-5} m, and their density was approximately $1.0 \times 10^3 \text{ kg m}^{-3}$. The motion of the particles was recorded with a Baumer high-speed camera. The field of view was located away from the boundaries of the tray. The main components of the experimental arrangement are shown in Fig. 1.

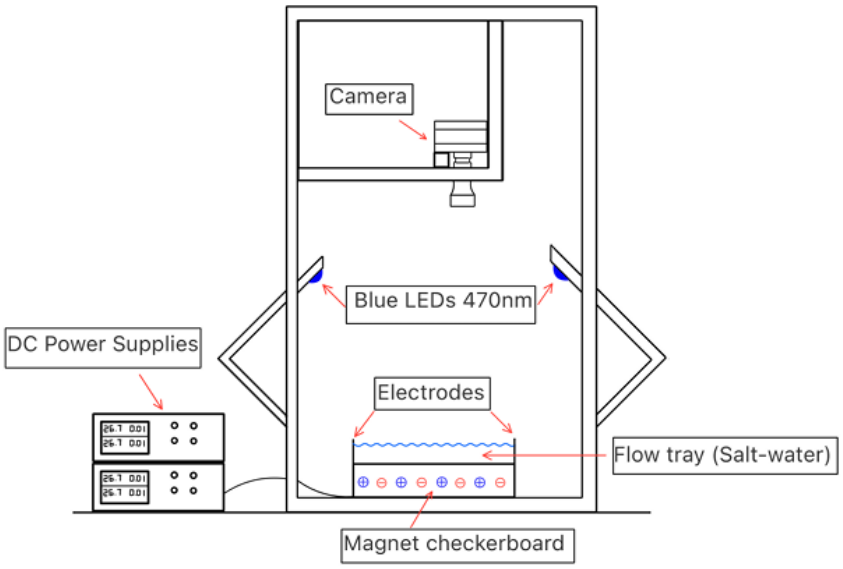


Figure 1: Experimental setup of the Q2D experiment

The camera observes the tracer particles from above. Two blue LED sources illuminate the fluid, allowing the fluorescent particles to be recorded as they move. The electrodes drive an electric current through the conductive layer, while the alternating-polarity magnets beneath the tray provide the magnetic field. The pixel-to-length conversion was obtained from the calibration image. A known distance of 20 mm on the calibration target corresponded to 231.18 px, giving

$$\Delta x_{\mathrm{px}} = \frac{20\ \mathrm{mm}}{231.18\ \mathrm{px}} = 0.086514\ \mathrm{mm\,px^{-1}} = 8.6514 \times 10^{-5}\ \mathrm{m\,px^{-1}}.$$

Table 1: Main parameters of the experimental setup. All dimensional quantities are given in SI units.

Parameter	Value	Unit
Flow tray length L_1	0.300	m
Flow tray width L_2	0.320	m
Fluid layer thickness h	5.0×10^{-3}	m
Magnet spacing L_f	3.5×10^{-2}	m
$CuSO_4$ mass fraction	0.10	–
Particle diameter d_p	7.5×10^{-5}–9.0×10^{-5}	m
Particle density ρ_p	1.0×10^{3}	kg m^{-3}
Nominal fluid density ρ_f	1.08×10^{3}	kg m^{-3}
Assumed kinematic viscosity ν	1.0×10^{-6}	m^2 s^{-1}
Pixel conversion Δx_{px}	8.6514×10^{-5}	m px^{-1}
Distance from boundaries	3.0×10^{-2}	m

2.2 Experimental conditions

Four forcing cases were recorded. The voltage was kept approximately constant at 28.2 V, while the current was varied. The frame rate was increased for the stronger forcing cases so that the particles did not move too far between two consecutive images. The exposure time was kept constant.

Table 2: Acquisition conditions for the four datasets.

Run	Current [A]	Voltage [V]	Frame rate [s^{-1}]	Exposure [s]	Frames [-]
ID1	0.010	28.2	10	3.0×10^{-3}	75
ID2	0.090	28.2	25	3.0×10^{-3}	187
ID3	0.250	28.2	50	3.0×10^{-3}	375
ID4	0.500	28.2	100	3.0×10^{-3}	747

2.3 Measurement validation

The acquisition settings were checked by estimating the particle displacement between two consecutive frames. This is important in PIV because the correlation becomes less reliable when particles move too far compared with the interrogation window. Using the measured RMS velocities, the calibration coefficient, and the frame rates in Table 2, the characteristic inter-frame displacements are 8.1, 13.1, 10.9, and 5.7 pixels for ID1 to ID4. All four values are below 16 pixels, or one quarter of the 64×64 pixel window used for the initial PIV pass. For the final 32×32 pixel window, the corresponding guideline is 8 pixels: ID1 is just above it, while ID2 and ID3 exceed it more substantially. The multi-pass procedure and vector validation help limit erroneous estimates, but these RMS-based checks do not establish that every local displacement satisfies the criterion. The frame rate and exposure time involve different constraints. Using the planning velocity of 40 $\mathrm{mm\,s^{-1}}$ and the calibrated pixel size, the one-quarter rule for a 32-pixel window would require approximately 58 $\mathrm{frames\,s^{-1}}$ at that speed. The recorded frame rates were instead adapted to the expected velocity of each run. With the measured RMS velocities and the fixed 3.0 ms exposure, the estimated particle travel during exposure ranges from about 21 μm (ID1) to 147 μm (ID4). This can exceed the nominal 80 μm tracer diameter at higher currents, so some motion blur is possible.

The tracer response provides another consistency check. Taking a representative diameter $d_p = 80$ μm, $\rho_p = 1000$ $\mathrm{kg\,m^{-3}}$, and an approximate dynamic viscosity $\mu = \rho_f \nu \simeq 1.08 \times 10^{-3}$ Pa s, the Stokes response time is

$$\tau_p = \frac{\rho_p d_p^2}{18\mu} \simeq 3.3 \times 10^{-4} \text{ s}.$$

Based on the forcing scale L_f and the planning velocity $U_{\mathrm{ref}} = 0.040$ $\mathrm{m\,s^{-1}}$,

$$Stk = \frac{\tau_p U_{\mathrm{ref}}}{L_f} \simeq 3.8 \times 10^{-4} \ll 1.$$

This small estimated Stokes number supports the use of the particles as flow tracers at the forcing scale; it does not rule out errors caused by image quality or unresolved smaller flow scales.

3 PIV Processing

The recorded images were processed with PIVlab [1], following the general PIV procedure introduced in the lecture notes [3]. Before the velocity calculation, the images were preprocessed to improve particle visibility. Background subtraction removed static reflections and non-moving features, while contrast enhancement made the tracer particles easier to detect. The velocity fields were computed using a multi-pass FFT cross-correlation method. The first pass used a 64×64 pixel interrogation window to estimate particle displacements. A second pass with 32×32 pixel windows was then used to improve the spatial resolution. After the correlation

step, the vectors were validated. Spurious vectors were removed using velocity limits and local median filtering, and missing values were interpolated. This reduced the influence of isolated wrong vectors on quantities such as vorticity and autocorrelation.

The final PIV fields give the horizontal velocity components, velocity magnitude, and vorticity for every frame. Representative fields are shown in Fig. 2.

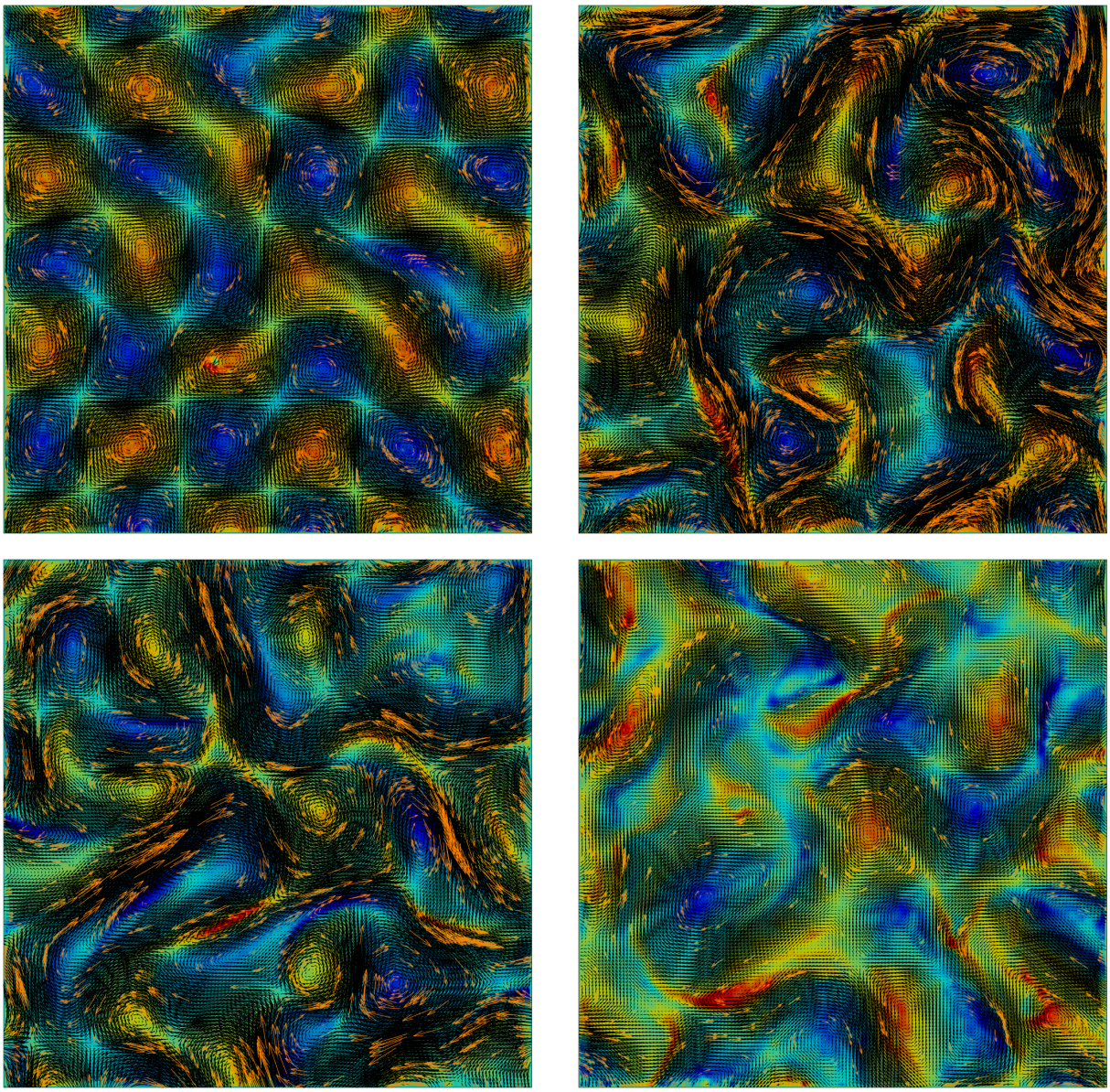


Figure 2: Representative processed PIV fields for the four forcing conditions. From left to right and top to bottom: ID1 at 0.010 A, ID2 at 0.090 A, ID3 at 0.250 A, and ID4 at 0.500 A. The colour fields represent vorticity and the arrows represent velocity vectors.

The PIVlab velocity output is given in pixels per frame. It was converted to SI units using

$$u = u_{\mathrm{px/frame}}\,\Delta x_{\mathrm{px}}\,f, \qquad v = v_{\mathrm{px/frame}}\,\Delta x_{\mathrm{px}}\,f,$$

where Δx_{px} is in $\mathrm{m\,px^{-1}}$ and f is the frame rate in $\mathrm{s^{-1}}$. The velocity magnitude is

$$|\mathbf{u}| = \sqrt{u^2 + v^2}.$$

The RMS velocity was computed from the velocity magnitude:

$$v_{\mathrm{RMS}} = \left\langle |\mathbf{u}|^2 \right\rangle^{1/2}.$$

The brackets denote an average over the field of view and over time. The Reynolds number was calculated with the magnet spacing as the characteristic length:

$$Re = \frac{v_{\mathrm{RMS}} L_f}{\nu}.$$

The kinematic viscosity was taken as

$$\nu = 1.0 \times 10^{-6}\ \mathrm{m^2\,s^{-1}}.$$

With v_{RMS} in $\mathrm{m\,s^{-1}}$, L_f in m, and ν in $\mathrm{m^2\,s^{-1}}$, the Reynolds number is dimensionless. The vorticity is defined as

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}.$$

The PIVlab vorticity output was converted to SI units using the frame rate, so that ω is expressed in $\mathrm{s^{-1}}$. The RMS vorticity was computed as

$$\omega_{\mathrm{RMS}} = \left\langle \omega^2 \right\rangle^{1/2}.$$

For the velocity statistics, the streamwise velocity was written as a mean plus a fluctuation. The normalized fluctuation was

$$q = \frac{u - \langle u \rangle}{\sqrt{\langle (u - \langle u \rangle)^2 \rangle}}.$$

This quantity is dimensionless. It was used for the probability density functions and for the kurtosis,

$$K = \left\langle q^4 \right\rangle.$$

A Gaussian distribution has a kurtosis of 3. The Eulerian velocity autocorrelation was calculated as

$$\Gamma_E(L) = \frac{\langle \mathbf{u}(\mathbf{x},t) \cdot \mathbf{u}(\mathbf{x}+\mathbf{r},t) \rangle_{|\mathbf{r}|=L,t}}{\langle \mathbf{u}(\mathbf{x},t) \cdot \mathbf{u}(\mathbf{x},t) \rangle_{\mathbf{x},t}}.$$

Here, $L = |\mathbf{r}|$ is the physical separation between two velocity vectors. In MATLAB, the two-dimensional vector autocorrelation is accumulated every fifth frame, accounting for valid velocity pairs, and then averaged over directions at fixed separation L. The result is plotted against L/L_f.

3.1 Uncertainties

The main uncertainties come from the current setting, the calibration, and the PIV processing. The current is set manually, so small deviations from the nominal value are possible. This matters most at the lowest current, where the relative error is larger. The calibration coefficient affects all dimensional velocity values. As a sensitivity estimate, an assumed ± 1-pixel uncertainty in the measured 231.18-pixel reference distance would correspond to a relative scale uncertainty of approximately 0.43%. The resulting relative uncertainty in v_{RMS} and Re due to calibration alone would be approximately the same. The ± 1-pixel assumption is illustrative, not a measured uncertainty bound. Normalized PDFs, kurtosis, and the normalized correlation amplitude are largely insensitive to a common velocity-scale factor, although the physical separation used for the autocorrelation still depends on the calibration. Other uncertainties may come from particle seeding, local particle clustering, reflections, or the removal of spurious vectors. These effects

can influence individual vectors, but the final quantities are averaged over many frames and over the full field of view.

Peak locking is another possible source of error. It can occur when the measured particle displacements are biased towards integer pixel values, especially if the sub-pixel estimation is not perfect. This effect was not quantified directly here, but the use of multi-pass processing, image deformation, and sub-pixel interpolation helps to reduce it.

4 Results and Discussion

The PIV analysis shows a clear increase in flow intensity with current, but the response becomes weaker at the highest forcing. The main numerical results are listed in Table 3.

Table 3: Processed PIV results for the four forcing conditions.

Run	Current [A]	v_{RMS} [m s^{-1}]	Re [-]	ω_{RMS} [s^{-1}]	K [-]
ID1	0.010	7.02×10^{-3}	246	0.866	2.24
ID2	0.090	2.84×10^{-2}	993	3.258	2.63
ID3	0.250	4.71×10^{-2}	1650	5.408	2.68
ID4	0.500	4.89×10^{-2}	1711	5.643	2.86

4.1 RMS velocity and Reynolds number

The Reynolds number follows the same evolution as the RMS velocity, since the characteristic length and the viscosity are constant. These two quantities therefore measure the increase in flow intensity.

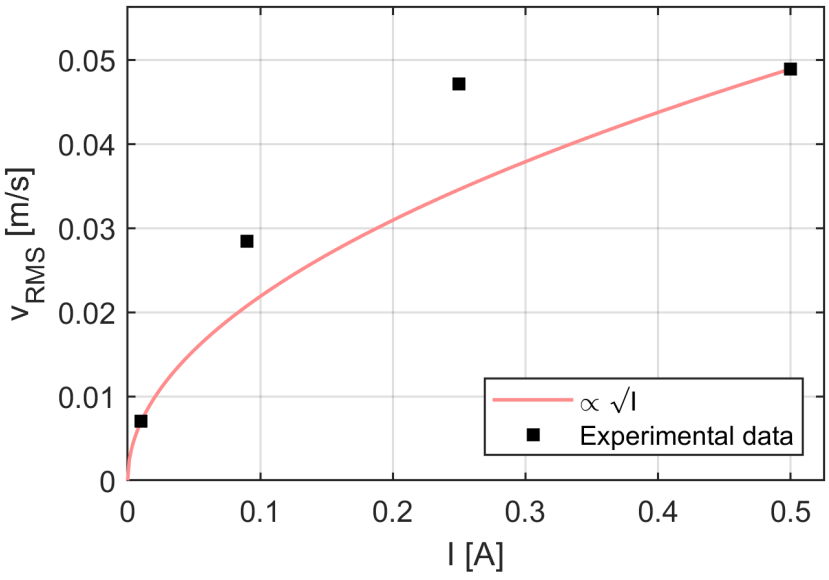


Figure 3: RMS velocity as a function of applied current. The red curve shows a reference square-root dependence with current.

The RMS velocity increases from 0.00702 m s^{-1} at 0.010 A to 0.0471 m s^{-1} at 0.250 A. However, doubling the current from 0.250 to 0.500 A only increases the RMS velocity by approximately 4%. This suggests that the velocity response may be approaching a plateau.

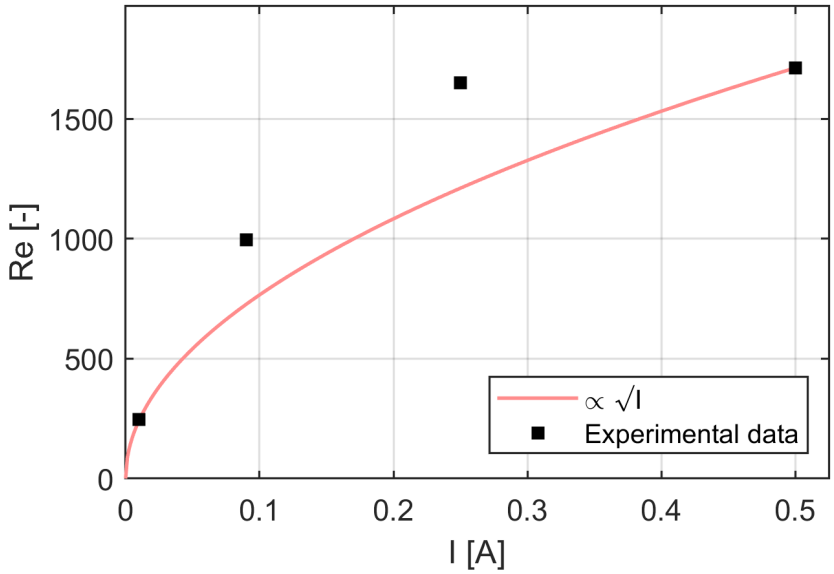


Figure 4: Reynolds number as a function of applied current. The characteristic length is the magnet spacing L_f.

The Reynolds number follows the same experimental trend as the RMS velocity (Fig. 4). In both plots, the red square-root curve is a reference passing through the highest-current data point, not a fit to all four measurements. The flattening between the two highest currents differs from the ideal $\sqrt{I}$ scaling; additional current settings would be needed to establish whether this is a true saturation effect.

4.2 RMS vorticity

The RMS vorticity gives the typical rotational strength of the flow. It also reflects the strength of the spatial velocity gradients.

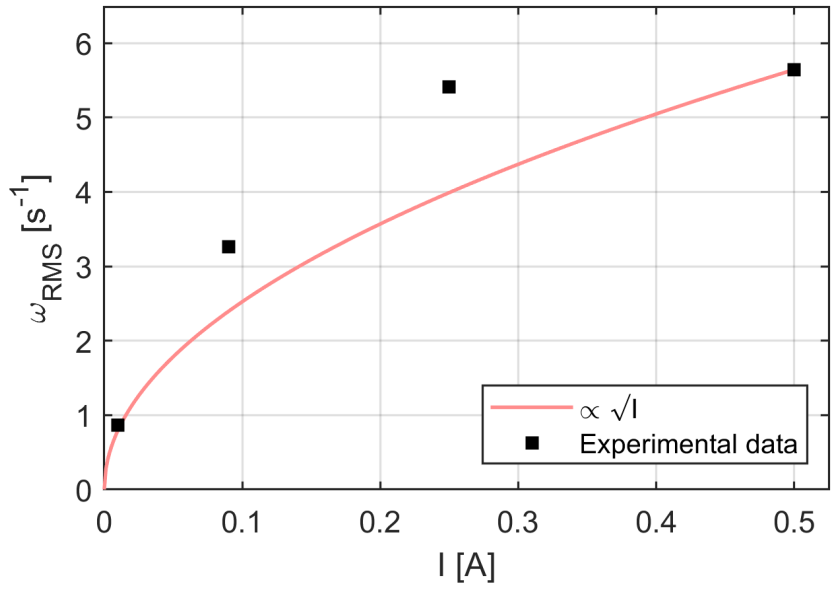


Figure 5: RMS vorticity as a function of applied current.

Figure 5 shows that the RMS vorticity increases from 0.866 to 5.643 s^{-1} across the four runs. As with the velocity, the increase is smaller between the two highest currents.

4.3 Eulerian velocity autocorrelation

The Eulerian velocity autocorrelation shows how much the velocity at one point is related to the velocity at another point separated by a distance L. It is therefore useful for identifying the spatial organisation of the flow. The autocorrelation curves are plotted against the physical separation normalized by L_f.

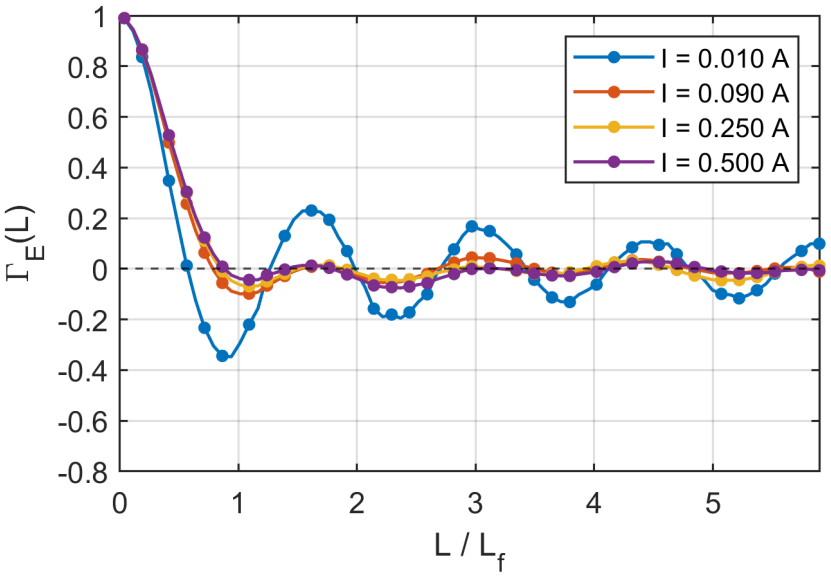


Figure 6: Eulerian velocity autocorrelation function as a function of normalized distance L/L_f. The distance L is the physical separation between two velocity vectors.

In Fig. 6, the correlation is close to one at small separations and can become negative at larger distances, indicating anticorrelated velocities. The lowest-current curve keeps a clearer oscillatory shape. This reflects the repeated vortex pattern created by the alternating magnets. At higher current, the oscillations are weaker and the correlation decays faster, showing that the velocity field loses coherence over shorter distances. This loss of spatial coherence is qualitatively consistent with the behaviour of quasi-two-dimensional turbulent flows discussed in the literature [4, 5]. However, the remaining oscillations indicate that the magnetic forcing pattern still influences the flow.

4.4 Probability density function of velocity fluctuations

The PDFs were calculated from the normalized streamwise velocity fluctuations and compared with a standard Gaussian distribution.

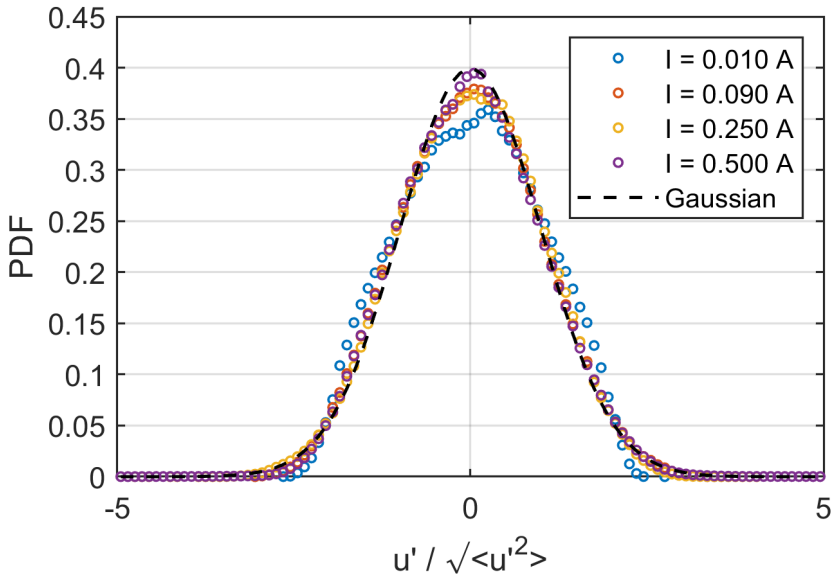


Figure 7: PDF of the normalized streamwise velocity fluctuations. The dashed black line is a Gaussian distribution.

The lowest-current distribution differs more from the Gaussian reference, while the higher-current PDFs move closer to it. The kurtosis values provide a numerical measure of this difference.

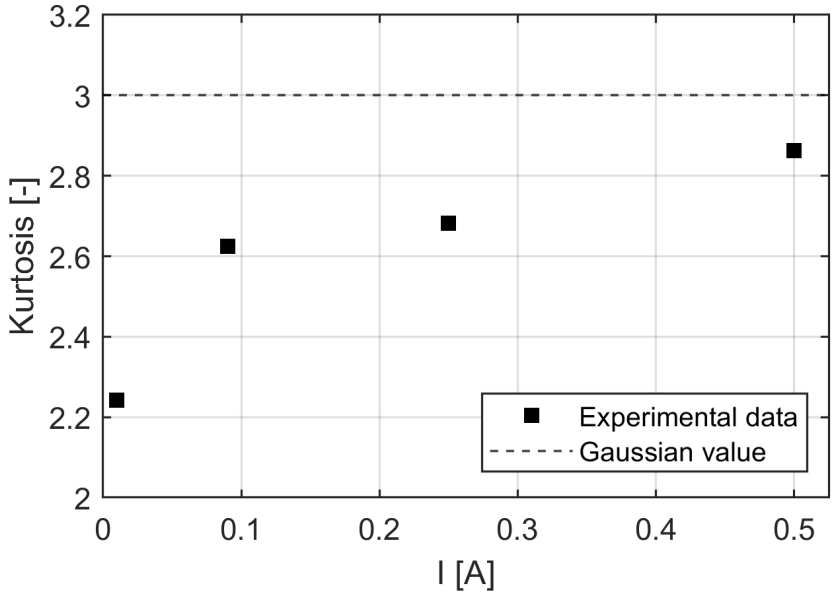


Figure 8: Kurtosis of the normalized streamwise velocity fluctuations. The dashed line shows the Gaussian value $K = 3$.

The kurtosis increases from 2.24 to 2.86, approaching the Gaussian value of 3 from below (Fig. 8). All four values are below 3, indicating negative excess kurtosis rather than positive excess kurtosis.

4.5 Transition towards a less coherent regime

The largest changes appear between ID1 and ID2, where the Reynolds number increases from 246 to 993 and the RMS vorticity from 0.866 to 3.258 s^{-1}. The weakening of the autocorrelation oscillations suggests that the magnet-induced flow pattern becomes less coherent as the forcing increases. With only four current values, the transition cannot be located precisely. The measurements suggest that the main change in flow organisation occurs between 0.010 and 0.090 A, but they do not establish a critical Reynolds number. Even at higher currents, the magnet forcing scale remains visible in the autocorrelation. The results therefore indicate a transition towards a less coherent quasi-two-dimensional regime, rather than providing sufficient evidence of fully developed turbulence.

5 Conclusions

This experiment studied an electromagnetically forced quasi-two-dimensional flow using Particle Image Velocimetry. The aim was to determine how the flow changes when the applied current is increased. The velocity, Reynolds number, and RMS vorticity all increase with current. The increase is strong at first, but becomes weaker between the two highest current values. This suggests that the response begins to saturate at the strongest forcing. The spatial statistics show that the imposed magnet pattern is important at low current. When the current increases, the autocorrelation oscillations become weaker and the velocity field loses coherence over shorter distances. The PDFs and kurtosis show a similar trend, with the velocity fluctuations moving closer to Gaussian statistics.

Overall, the experiment captures the evolution from a magnetically organised flow towards a less coherent quasi-two-dimensional regime. The strongest change appears to occur between the two lowest current values, but more measurements in this range would be needed to locate the transition more accurately.

References

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